

How we got our numbers

Every figure in our pitch, traced from the raw public data, through each calculation, to the slide.

Our pitch rests on two claims. First, that mobile-tower links can measure rain well enough to pay people on. Second, that Kavach can be priced so it survives bad years. This document shows exactly how we tested both: which data, which steps, which formulas, and what the results do and don't prove.

You don't need to be technical. Each part starts with the idea in plain words, then goes one level deeper. Every number can be re-checked in the Excel workbook or the Test Lab.

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Three chains of evidence

Every number belongs to one of three chains. Each chain starts with public data and ends with a claim we make on stage.

1 • Can towers see rain?

500 real tower links in Germany → our software turns signal loss into rain → compared with radar → **correlation 0.95.**

2 • Can we pay on it?

Our rain map → a pay/no-pay decision per square per day → compared with radar's decision → **94.6% agree, 68% caught, 18% false.**

3 • Can we price it?

25 years of IMD records → how often each trigger fires → **price per month** → replay every year → **stress test** → business case.

Where every number comes from

NUMBER	WHAT IT MEANS	BUILT FROM	CHECK IT
0.95	Network-average rain, ours vs radar, hour by hour, on unseen days	500 links, 192 test hours	Workbook · Network hourly
0.88 · 91%	Typical single link; share of links above 0.7	490 links scored one by one	Workbook · Link scores
0.78	Our street map vs radar map, every square, every hour	1,782 squares × 192 hours	Test Lab · tab A
×1.17	The one scale factor, fitted on tuning days only	Totals on 10–12 May	Workbook · Calibration
94.6% · 68% · 18%	Decisions that agree · radar events caught · our triggers not confirmed	14,256 square-days	Workbook · Payout test
11.9 days	Heavy-rain days a year in Mumbai (64.5 mm+)	IMD Santacruz, 2000–2024	Workbook · Prices
₹208 · ₹125	Kavach Plus per rider per month (5-day cap · 3-day cap)	Days paid × ₹300 ÷ 0.6 ÷ 12	Workbook · Prices
₹45 · ₹52 · ₹27→₹29	Kavach Basic: Mumbai rain · Nagpur heat · Ahmedabad heat (cost, then floor)	Same formula, rarer triggers	Workbook · Prices
222% · 242% · 469%	Worst year's claims ÷ premiums for each Basic product	Every year replayed	Workbook · Stress test
₹12.5 Cr · ₹174 Cr	Premiums a year: 50,000 riders · 1% of Jio's users	Users × price × 12	Workbook · Business case

THE HONEST FRAMING

Chains 1 and 2 use German data because no Indian telecom publishes link signals. They prove the **method** works on a real network. Chain 3 uses Indian data and proves the **prices**. Phase 1 of our plan repeats chain 1 on Jio's own Mumbai links.

The German data: what is in the files

A microwave link is a radio beam between two towers that carries phone and internet traffic. Every link logs how strong its signal is. When rain falls between the towers, the signal weakens. That log is our rain gauge.

Where it comes from

The **pycomlink example dataset**. pycomlink is the open-source Python toolbox for rain from microwave links, developed at Karlsruhe Institute of Technology (KIT) in Germany. Its code and data are public on GitHub (github.com/pycomlink/pycomlink, folder **pycomlink/io/example_data**) under a BSD licence. The three files we used are byte-for-byte identical to the copies there.

FILE	WHAT IS INSIDE	SIZE
<code>example_cml_data.nc</code>	500 links × 2 channels × 15,840 minutes (10–20 May 2018). For each minute: transmitted and received signal level in dBm. For each link: frequency, polarization, length, both tower positions.	5.5 MB
<code>example_path_averaged_reference_data.nc</code>	German Weather Service (DWD) radar rainfall averaged along each link's path, every 5 minutes. The answer key for each link.	2.4 MB
<code>example_areal_reference_data.nc</code>	The radar map itself (RADOLAN-YW), a 190 × 228 grid of about 1 km squares, every 5 minutes. The answer key for the street map.	10.2 MB

What one link looks like

LINK 359	VALUE	MEANING
Length	8.47 km	Distance between the two towers
Frequency	23.1 GHz	Higher frequencies are more sensitive to rain
Transmitted (dry minute)	19.0 dBm	Power sent
Received (dry minute)	about -40.6 dBm	Power arriving: the beam loses about 59.6 dB on a dry day
Received (22:45, 13 May)	-66.1 dBm	During heavy rain: about 25 dB weaker

WHY GERMANY

It is the only public dataset we could find with real telecom link signals and a radar answer key for the same minutes. Indian operators do not publish link data. That is exactly what Phase 1 asks Jio for.

ONE QUIRK TO KNOW

The tower coordinates are shifted for privacy: the file places the network at 57–58°N, 1–3°E, which is in the North Sea. Distances and shapes are real; the location is not. Radar was shifted the same way, so comparisons still line up.

Why rain weakens the signal, in one page

Raindrops absorb and scatter microwaves. The more rain between two towers, the more of the beam is lost. Engineers have measured this relationship for decades, so it has a standard formula.

Decibels, briefly

Signal strength is measured in **decibels (dB)**, a scale where every 10 dB means ten times less power. dBm is the same scale measured against one milliwatt. "Signal loss" here is simply **sent minus received**: $19.0 - (-40.6) = 59.6$ dB on a dry minute for link 359.

The standard formula (ITU-R P.838)

$$k = a \times R^b$$

k = loss per kilometre (dB/km) · R = rain rate (mm per hour) · a, b = constants for the link's frequency and polarization

The International Telecommunication Union publishes a and b for every frequency. Turned around, the formula gives rain from loss: $R = (k \div a)^{1/b}$. For link 359 (23.1 GHz, vertical), $a = 0.1294$ and $b = 0.9624$.

RAIN RATE	LOSS PER KM AT 23.1 GHZ	LOSS OVER LINK 359'S 8.5 KM	WHAT IT FEELS LIKE
1 mm/h	0.13 dB/km	1.1 dB	Light drizzle: barely visible in the signal
10 mm/h	1.19 dB/km	10.0 dB	Heavy shower: the signal drops ten-fold
25 mm/h	2.87 dB/km	24.3 dB	Downpour: what link 359 saw at 22:45 on 13 May

WHY FREQUENCY MATTERS

Raindrops are a few millimetres across. Waves of similar size (above about 10 GHz) interact with them strongly. The dataset's links run from 6.5 to 38.9 GHz; most sit between 15 and 40 GHz, the sweet spot for sensing rain.

WHY IT MEASURES A LINE

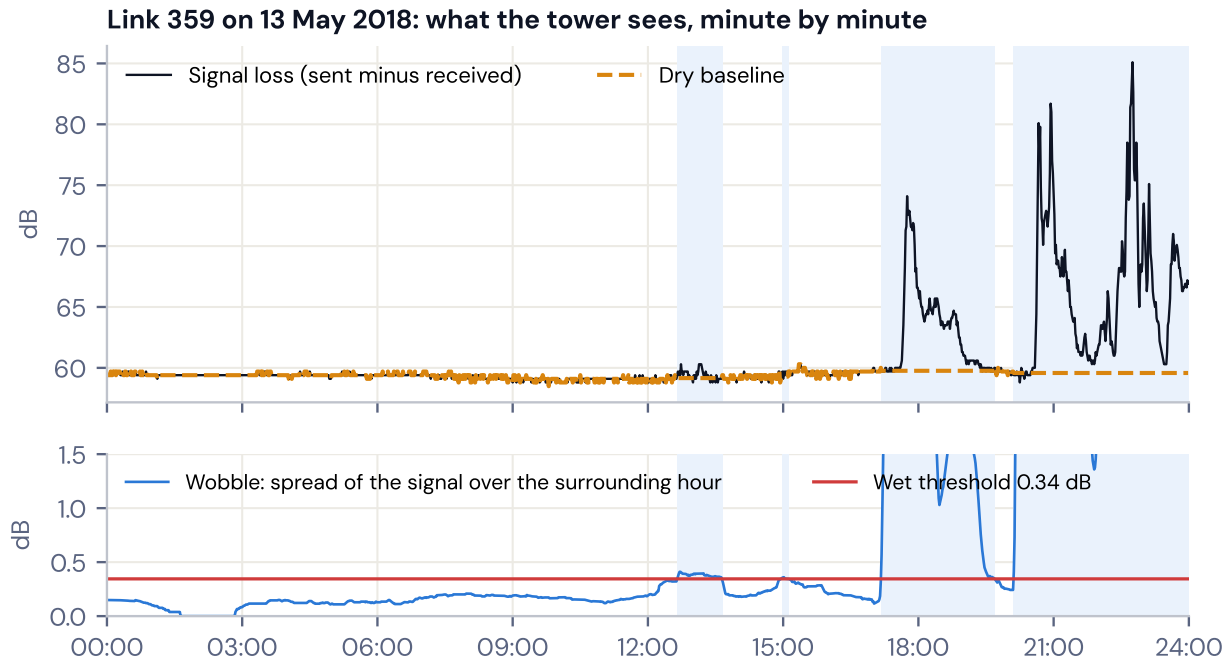
A link reports the total loss along its whole path. So it tells you the **average rain along a road**, not at one house. Many crossing links let you rebuild the pattern street by street.

TWO THINGS THAT ALSO WEAKEN THE SIGNAL

Water on the antenna (a wet dish loses a little signal even after rain stops) and **everyday wobble** (temperature, humidity, equipment). The pipeline on the next pages removes both before turning loss into rain.

Steps 1–3: from raw signal to "it is raining"

Our program, `rain_pipeline.py`, runs the same seven steps on every link and channel. Here is link 359 on the stormy day of 13 May 2018.



Top: the signal loss every minute (black) and the dry baseline the code holds during rain (dashed). Bottom: the wobble, and the threshold above which a minute counts as wet. Blue bands are the minutes marked wet: 453 of 1,440 that day.

- 1 Read the signal.** Loss = transmitted – received, every minute. Missing readings are stored as 255 (sent) or –99.9 (received) and are treated as gaps.
- 2 Fill short gaps.** Gaps of up to 5 minutes are bridged with a straight line. Longer gaps stay empty.
- 3 Spot rain from the wobble.** For each minute, take the standard deviation of the signal over the surrounding 60 minutes. On dry days the signal barely moves; rain makes it jump. The threshold is set per channel: **$1.12 \times \text{that channel's 80th-percentile wobble}$** . For link 359 that is 0.34 dB. At 22:45 the wobble was 5.9 dB, far above it.

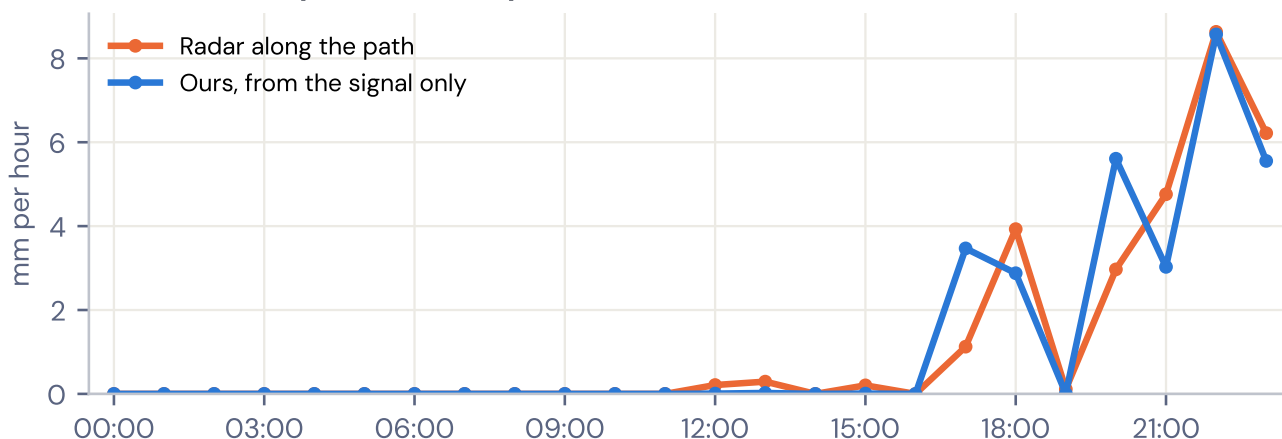
IN PLAIN WORDS

Every link learns what "normal jitter" looks like for itself. When the signal shakes much more than usual, it is raining on that link. The 80th percentile means the link is assumed dry most of the time, which is true almost everywhere.

Steps 4–7: from "it is raining" to millimetres

- 4 Hold a dry baseline.** Outside rain, the baseline follows the signal. When a wet spell starts, it freezes at the average of the last 5 dry minutes and stays there until the spell ends. Anything above it during rain is extra loss.
- 5 Remove the wet-antenna effect.** Water on the antennas causes extra loss that is not rain in the path. Following Schleiss et al. (2013), the code allows up to **1.5 dB** for it, building up over about 15 minutes of rain.
- 6 Turn loss into rain.** Rain loss = signal loss – baseline – wet-antenna allowance. Divide by length for dB/km, then apply $R = (k \div a)^{1/b}$. Rates below 0.1 mm/h are set to zero. The two channels of each link are averaged, then averaged into hours.
- 7 Apply one scale factor.** Multiply every link by **K = 1.173**, fitted on the tuning days only (page 8).

Link 359, hourly rain on 13 May: our estimate vs radar



Link 359's hourly rain on 13 May: our estimate (from the signal alone) against radar along the same path. Day total: ours 29.1 mm, radar 28.5 mm.

SETTING	VALUE	WHY THIS VALUE
Wobble window	60 minutes	Long enough to see a storm build, short enough to catch it starting
Wet threshold	1.12 × 80th percentile	Standard rolling-deviation method; 1.12 balances missed and false wet minutes
Baseline memory	last 5 dry minutes	Avoids freezing on a single noisy reading
Wet-antenna cap	1.5 dB	Typical value in the published research for links like these

WHAT IS OURS AND WHAT IS BORROWED

Steps 4–6 use pycomlink's published functions, the peer-reviewed standard. The wet/dry rule, the calibration, the scoring, the street map and the payout test are our own code. In the Test Lab you can move the threshold and wet-antenna sliders and watch every result change.

One minute, calculated by hand

Link 359, channel 1, 13 May 2018 at 22:45 UTC: the heaviest minute of the storm on this link. Every number below comes from the data file and can be found in `three_links_minute_by_minute.csv`.

STEP	CALCULATION	RESULT
1 · Signal loss	$19.0 - (-66.1)$	85.1 dB
3 · Is it raining?	Wobble over the surrounding hour 5.91 dB, threshold 0.34 dB	Yes, wet
4 · Dry baseline	Frozen when this spell began (average of the last 5 dry minutes)	59.58 dB
4 · Extra loss	$85.10 - 59.58$	25.52 dB
5 · Wet antenna	Built up to its cap during the storm	- 1.50 dB
6 · Rain loss	$25.52 - 1.50$	24.02 dB
6 · Loss per km	$24.02 \div 8.468 \text{ km}$	2.837 dB/km
6 · Rain rate	$(2.837 \div 0.1294)^{1 \div 0.9624} = 21.92^{1.0391}$	24.7 mm/h
6 · Both channels	Channel 2 gave a slightly lower rate; the link uses the average	24.1 mm/h
6 · The hour	Average of the 60 one-minute values (both channels), 22:00 to 22:59	7.31 mm/h
7 · Calibrated hour	7.31×1.173	8.57 mm/h
Answer key	Radar along the same path, same hour	8.63 mm/h

WHAT THIS SHOWS

A handful of subtractions and one standard formula turn a phone network's own log into rain within 1% of radar for this hour. No new hardware, no guesswork.

NOT EVERY HOUR IS THIS CLOSE

One hour earlier the same link read 3.0 mm/h against radar's 4.8. Single hours are noisy; that is why we judge on 192 unseen hours and 490 links, not on one good example.

SAY IT LIKE THIS

"At 22:45 the signal on one link dropped 25 dB. Take off 1.5 for the wet antenna, apply the ITU formula for 23 GHz, and you get 25 mm an hour. Radar saw the same storm."

Tune on three days, test on eight

A test is only honest if the answer key is hidden while you set things up. We split the 11 days in two and never let the scoring days influence the method.

PERIOD	DAYS	USED FOR
Tuning	10–12 May 2018 (72 hours)	Fitting the one scale factor K. Nothing else.
Test	13–20 May 2018 (192 hours)	Every published score: 0.95, 0.88, 0.78, 94.6%, 68%, 18%

The scale factor K, step by step

Our raw estimates read a little low (wet-antenna and baseline choices are conservative). K scales them so that, over the tuning days, our total rain equals radar's total.

Radar: all links, all tuning hours, added up	2,758.0
Ours, before scaling: same sum	2,351.9
K = radar ÷ ours	1.173

Three things people get wrong about K

0.95

K cannot create the correlation. Multiplying every value by the same number changes the size, not the up-and-down pattern. The network correlation is 0.9531 with K and exactly the same without it. The pattern was in the raw signal.

1

It is one number for the whole network, not one per link or per day. With 500 links and 192 test hours, a single number cannot be tuned to fit the answers.

0.82

It does not make totals perfect. On the test days our rain still adds up to 82% of radar's (the "bias"). K fixes most of the gap; local calibration in the pilot fixes the rest.

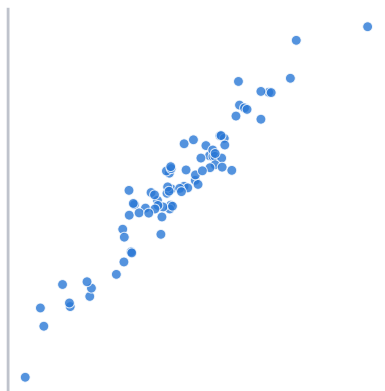
WHERE K DOES MATTER

In the payout test. A trigger is a fixed amount of rain (10 mm in 3 hours), so reading 18% low would mean missing payouts. That is why the pilot calibrates against Mumbai's own rain gauges continuously, not once.

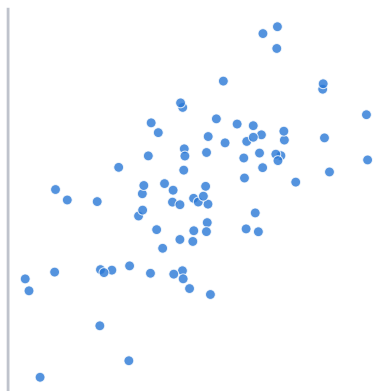
What a correlation of 0.95 actually means

Correlation (written r) measures how closely two series rise and fall together. It runs from -1 to 1 . At 1 , every rise in one is matched by a rise in the other. At 0 , there is no pattern at all.

$r \approx 0.95$ (real r 0.96)



$r \approx 0.6$ (real r 0.62)



$r \approx 0$ (real r 0.07)

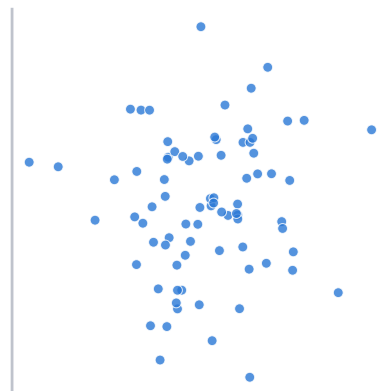


Illustration with made-up points: what different correlations look like. Each dot pairs two measurements of the same moment.

The formula, and a tiny example from our data

$r = \frac{\sum(dx \times dy)}{\sqrt{(\sum dx^2 \times \sum dy^2)}}$, where dx and dy are each value's distance from its own average. Six real hours of network-average rain on 13 May:

HOUR	OURS (MM/H)	RADAR (MM/H)	DX	DY	DX × DY
13:00	0.220	0.396	-0.614	-0.642	0.3947
14:00	0.427	0.538	-0.407	-0.500	0.2039
15:00	0.820	1.218	-0.014	+0.179	-0.0026
16:00	0.993	1.353	+0.159	+0.315	0.0499
17:00	1.112	1.184	+0.278	+0.145	0.0404
18:00	1.434	1.542	+0.600	+0.504	0.3019
Averages: ours 0.834, radar 1.038			$\sum dx^2$ 1.0054	$\sum dy^2$ 1.0691	$\sum$ 0.9882

$r = 0.9882 \div \sqrt{(1.0054 \times 1.0691)} = \mathbf{0.95}$. The same calculation over all 192 test hours gives **0.953**. In Excel it is one function: `=CORREL(ours, radar)`.

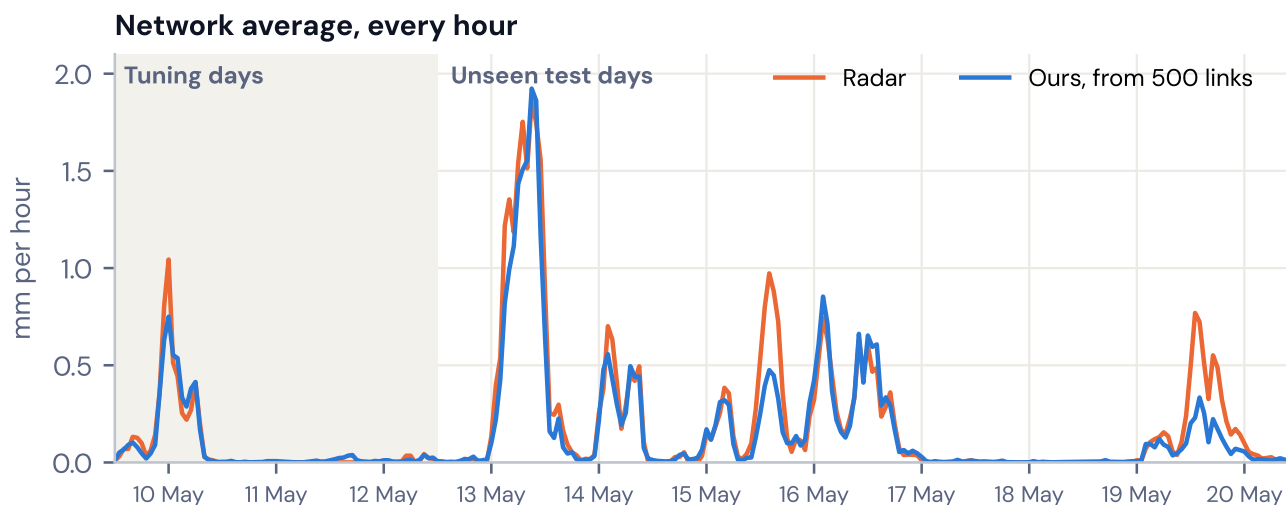
WHAT IT DOES TELL YOU

Whether our system sees rain arrive, build, peak and leave at the same times and places as radar. That is what a payout trigger needs.

WHAT IT DOESN'T

Whether the amounts are right. A system reading exactly half of radar every hour would still score 1.0. That is why we also report the bias (0.82) and the payout test.

Our four correlations, and why they differ



Average rain over all 500 links, every hour for 11 days. Shaded: tuning days. Both lines come from completely different instruments: tower signals and weather radar.

COMPARISON	R	WHAT IS BEING COMPARED	WHY IT IS HIGHER OR LOWER
Network average	0.953	The average over 500 links, each test hour (192 pairs)	Averaging cancels each link's random errors. Highest.
Single link (median)	0.882	Each link vs radar on its own path; the middle of 490 scores	One link's noise is not averaged away. 91% of links still score above 0.7.
Every link-hour pooled	0.801	All 96,000 link-hour pairs thrown together	Mixes good and weak links and every quiet hour.
Street map vs radar map	0.779	Each 4 km square, each test hour	Adds map-making error: squares far from links must be filled in. Lowest.

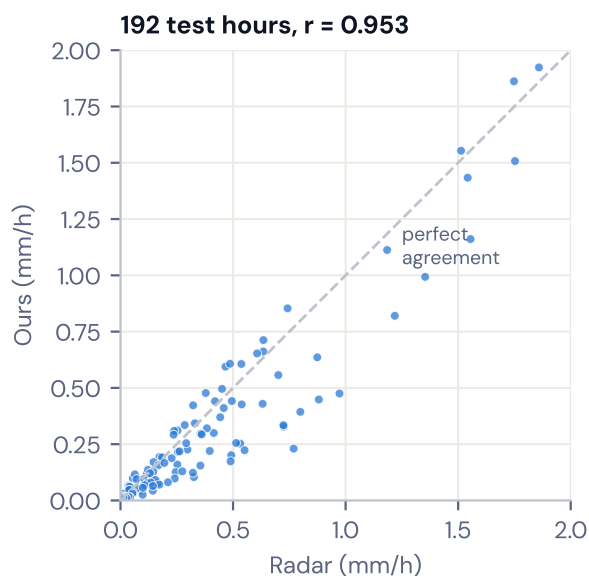
AN ANALOGY

Ask 500 people to guess the temperature. Each guess is a bit off, but the average is excellent. The network number (0.95) is the average; the single-link number (0.88) is a typical individual; the map (0.78) also asks people to guess for streets they cannot see.

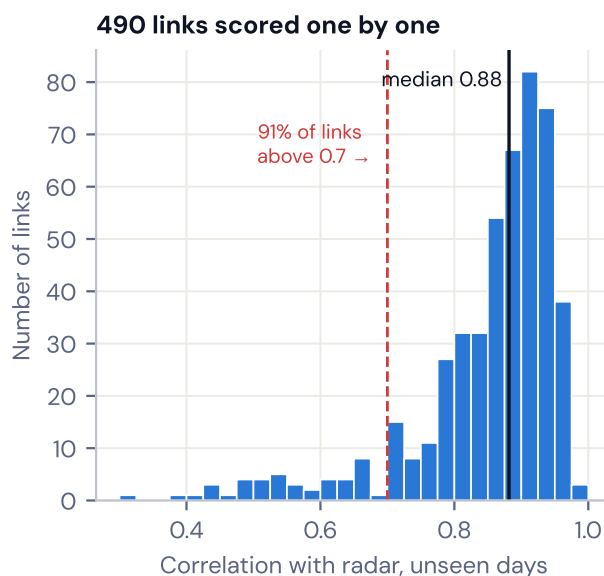
SAY IT LIKE THIS

"Averaged across the network we match radar at 0.95. A typical single link scores 0.88, and nine in ten links are above 0.7. The street map is 0.78 because it has to fill in gaps, and that is what fusion with radar and gauges improves."

Looking closer: every hour and every link



Each dot is one test hour. Dots on the dashed line would be perfect agreement. They cluster tightly along it, sitting a little below: the 18% under-reading.



Each link scored on its own. Most links sit between 0.8 and 0.95. A long tail of weaker links has no single cause.

What separates strong links from weak ones

LINK	R	FREQUENCY	LENGTH	WHY
Best · 118	0.99	37.4 GHz	2.6 km	Short path at a high frequency (37 GHz) that reacts strongly to rain
Typical · 359	0.88	23.1 GHz	8.5 km	The example used throughout this document
Weak · 33	0.70	6.5 GHz	28.6 km	Low frequency (6.5 GHz) barely reacts to rain; over 28.6 km other losses get counted as rain. It read 165 mm against radar's 78

Only 10 of 500 links were not scored: they saw too little rain on the test days (under 5 mm in total) or had too few valid hours for a fair score.

Across all links, length and frequency barely predict the score (correlation about 0.02 with each), and weak links are the same typical length (about 6 km) and frequency (about 25 GHz) as strong ones. They did see a little less rain on the test days (median 38 mm against 43 mm), so noise weighed more. Link 33 is the exception that shows a real rule: very low frequencies are poor rain sensors.

WHY THIS MATTERS FOR INDIA

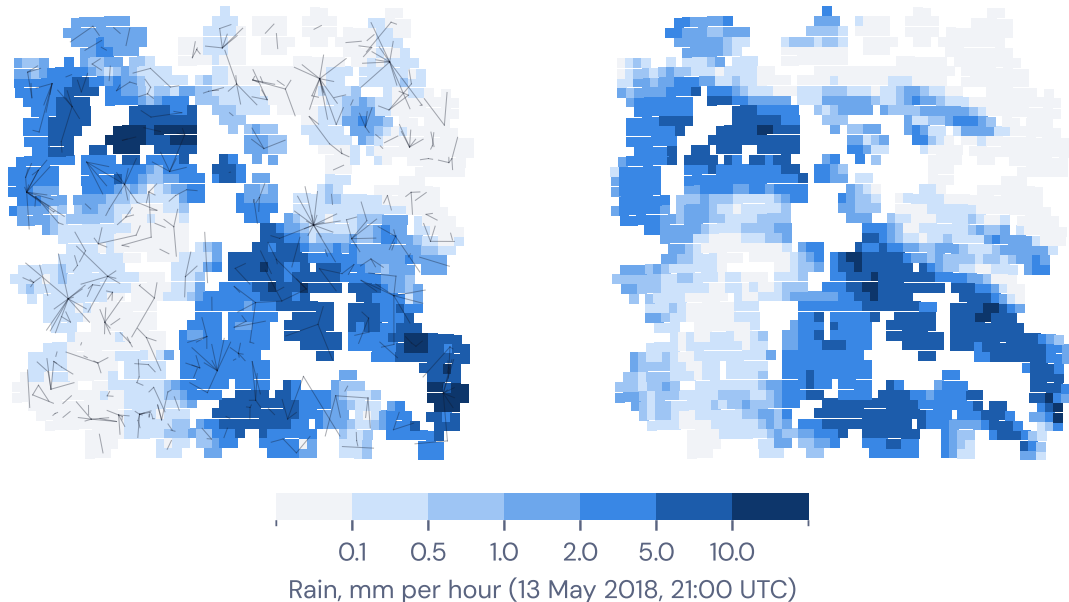
Monsoon rain is heavier than a German May, so signal drops are bigger relative to noise. But link quality has to be measured, not assumed, which is why Phase 1 starts with a coverage audit of Jio's Mumbai links: frequency, length and density in every square, and each link's own score against gauges.

From 500 lines to a map of squares

Payouts happen per area, so we need rain for every square of the city, not just along links.

Our map, from 500 links only

Radar (the answer key)



The heaviest test hour (13 May, 21:00 UTC). Left: built only from the 500 links (thin lines). Right: radar. Squares are about 3 × 4 km (radar's 1 km grid, four by four).

How each square gets its value

- 1 Each link's hourly rain is placed at the **midpoint** of its path.
- 2 For every square, find the **8 nearest midpoints** within 20 km.
- 3 Average them, giving nearer links more say: weight = $1 \div \text{distance}^2$ (inverse distance weighting). A link 2 km away counts four times as much as one 4 km away.
- 4 Only squares with a link midpoint within 6 km are scored: **1,782 squares**. Elsewhere we say "not enough links" instead of guessing.

THE SIMPLIFICATION

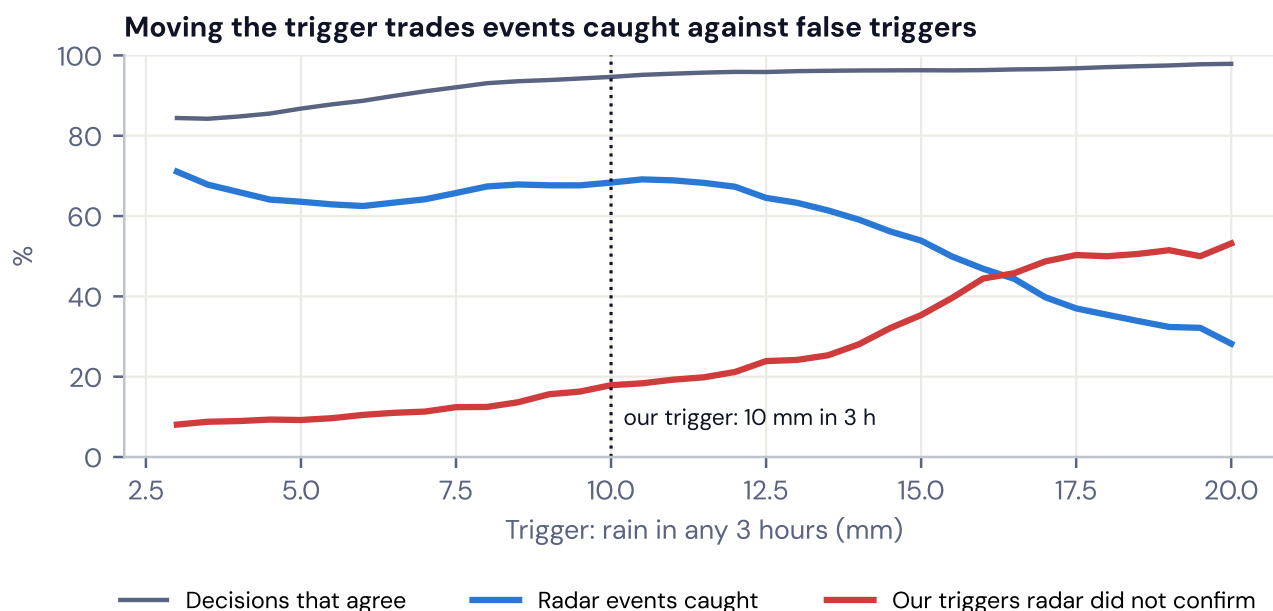
Putting a link's rain at its midpoint ignores that it measured a whole line. Production methods treat each reading as an average along the line and blend in radar and gauges. Our simple version still reaches 0.78 against radar, which is a floor to build on, not a ceiling.

Would we have paid the right squares?

Correlation is about patterns. Insurance is about decisions. So we asked the question a customer would: on each day, in each square, would our system and radar both have said "pay"?

The rule: a square pays on a day if any 3-hour window that day had at least **10 mm** of rain. We apply it twice, once to our map and once to radar's, for 1,782 squares × 8 test days = **14,256 decisions**.

	RADAR: PAY	RADAR: DON'T	Agreement	$(1,120 + 12,373) \div 14,256$	94.6%
Ours: pay	1,120 both pay	244 false trigger			
Ours: don't	519 missed	12,373 neither	Events caught	$1,120 \div (1,120 + 519)$	68.3%
			False triggers	$244 \div (1,120 + 244)$	17.9%



Every trigger level from 3 to 20 mm, re-scored. Lower triggers catch more events but agreement falls; higher triggers cut events caught and raise false triggers. 10 mm sits near the best balance.

WHY 94.6% FLATTERS US

Most square-days are dry, and "neither pays" counts as agreement. Always say 68% and 18% alongside it. They are the numbers an insurer cares about.

HOW THE PILOT CLOSSES THE GAP

Blend links with IMD radar and gauges, calibrate locally, and set the target: 80%+ of events caught, under 15% false triggers, before money moves.

25 years of IMD records

To price Kavach we need to know how often each trigger would have fired in each city. That needs long, daily, official records.

Source

IMD's synoptic station reports, as archived in NOAA's **Global Summary of the Day**, a public-domain collection of daily reports from weather stations worldwide (noaa-gsod-pds.s3.amazonaws.com). We used IMD's own website first, but it was not reachable from our environment; the NOAA archive carries the same station reports.

CITY	STATION	STATION ID	DAYS KEPT	DAYS WITHOUT RAIN REPORT
Mumbai	Santacruz airport	43003	9,119	150
Nagpur	Sonegaon airport	42867	8,932	778
Ahmedabad	Ahmedabad airport	42647	9,052	188
Mumbai (2nd)	Colaba	43057	Used only to show how different two stations 20 km apart can be	

Cleaning, step by step

- 1 Convert units.** Rain comes in inches: $\times 25.4$ gives mm (2.54 in = 64.5 mm). Temperature comes in °F: $(^{\circ}\text{F} - 32) \times 5 \div 9$ gives °C ($116.6^{\circ}\text{F} = 47.0^{\circ}\text{C}$).
- 2 Drop missing codes.** 99.99 means "no rain report" and 9999.9 means "no temperature". Those days count as no trigger, never as a trigger.
- 3 Keep only complete years.** A year counts only with at least 330 daily reports and at least 110 in June–September, so a gap in the monsoon can't make a year look dry. All 25 years (2000–2024) passed for all three cities.

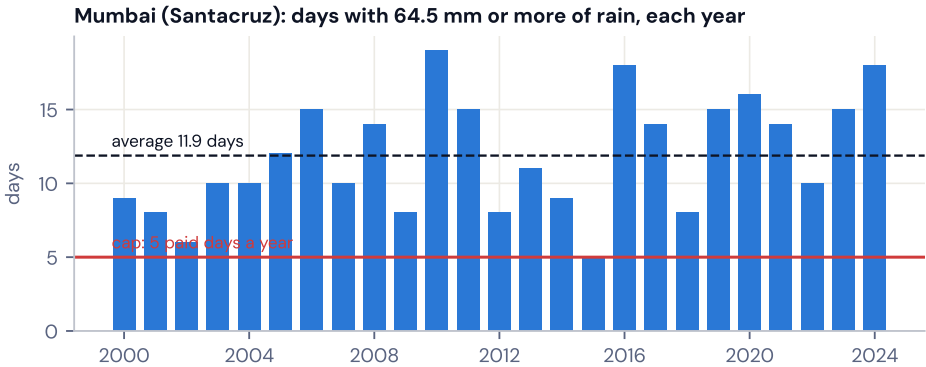
A KNOWN WEAKNESS OF THE ARCHIVE

It can under-record single extreme days. For 26 July 2005, IMD's official Santacruz total is 944 mm; the archive shows 461 mm. Both are far above every trigger, so the day still counts. We use the archive to count how often thresholds are crossed, never for exact totals.

How often would Kavach have paid?

We use IMD's own definitions as triggers, so nobody can say we picked numbers to suit ourselves.

IMD's terms, for one day: heavy rain 64.5–115.5 mm · very heavy 115.6–204.4 mm · extremely heavy 204.5 mm or more · heatwave at an actual 45°C or more, **severe heatwave** at 47°C or more.



Every year since 2000, Santacruz had at least 5 heavy-rain days (fewest: 5 in 2015; most: 19 in 2010). So a 5-day cap is reached every single year.

CITY	TRIGGER	DAYS A YEAR (AVG)	PAID, 5-DAY CAP	YEARS WITH ANY	COST PRICE A MONTH
Mumbai	Heavy rain, 64.5 mm+	11.9	5.0	25 of 25	₹208
Mumbai	Very heavy, 115.6 mm+	4.4	4.0	25 of 25	₹165
Mumbai	Extremely heavy, 204.5 mm+	1.1	1.1	15 of 25	₹45
Nagpur	45°C+	9.7	4.0	23 of 25	₹168
Nagpur	47°C+ (severe heatwave)	2.2	1.2	10 of 25	₹52
Ahmedabad	45°C+	5.4	2.4	15 of 25	₹100
Ahmedabad	47°C+ (severe heatwave)	0.7	0.6	6 of 25	₹27

WHY THESE TRIGGERS FOR THESE PRODUCTS

Kavach Plus (riders) uses heavy rain, 64.5 mm: the level at which deliveries stop. It fires often, so it behaves like a predictable rain allowance. **Kavach Basic (everyone)** uses the rare extremes, 204.5 mm in Mumbai and 47°C in Nagpur and Ahmedabad: true disaster days, about one a year, which keeps it cheap. Mumbai's hottest day in these records was 42°C, far below the heat triggers, so heat cover starts in Nagpur and Ahmedabad.

From trigger days to a monthly price

$$\text{Price per month} = \text{average days paid a year} \times \text{payout per day} \div \text{payout share} \div 12$$

Kavach Plus, worked through

Heavy-rain days a year in Mumbai (average of 25 years)	11.88
Capped at 5 paid days (every year reaches the cap)	5.00
Expected claims per rider a year: $5 \times ₹300$	₹1,500
Premium a year so claims are 60% of it: $₹1,500 \div 0.6$	₹2,500
Per month: $₹2,500 \div 12$	₹208

PRODUCT	TRIGGER	DAYS PAID A YEAR	COST PRICE	PRICE CHARGED
Kavach Plus · Mumbai riders	64.5 mm, cap 5	5.00	₹208	₹208
Kavach Plus · 3-day cap	64.5 mm, cap 3	3.00	₹125	₹125
Kavach Basic · Mumbai	204.5 mm	1.08	₹45	₹45
Kavach Basic · Nagpur	47°C	1.24	₹52	₹52
Kavach Basic · Ahmedabad	47°C	0.64	₹27	₹29 (floor)

The three choices behind the formula

₹300

Payout per trigger day. Roughly a day's lost earnings for a gig worker or vendor, large enough to matter, small enough to keep premiums low.

60%

Payout share (loss ratio). Of every ₹100 of premium, ₹60 goes back as claims. The other ₹40 pays the insurer's costs, reinsurance, Jio's distribution fee (15–25%) and margin. It is a design choice, and the actuaries confirm it before launch.

5 days

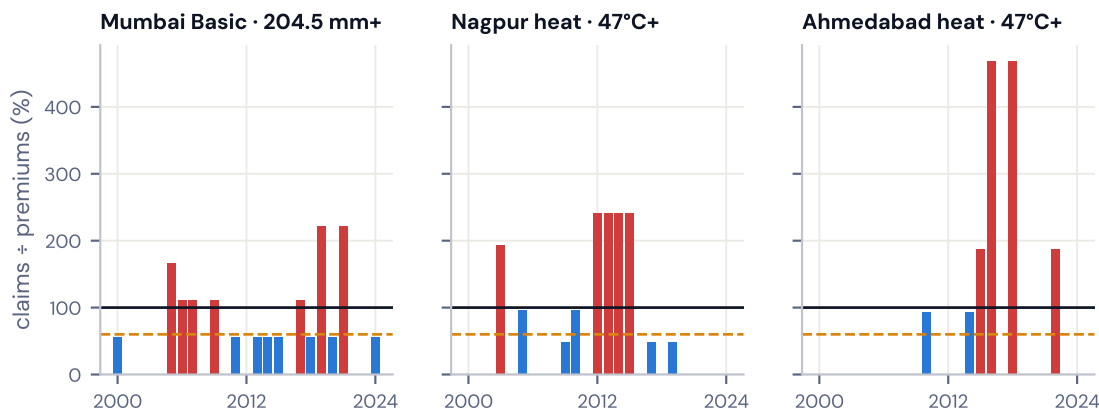
Yearly cap. Limits what one person can get in a year to ₹1,500, which makes the worst case known in advance. Fewer days (3) makes it cheaper; more days makes it richer but costlier.

WHERE ₹29 COMES FROM

Our original design rule: a typical location should trigger about 0.7 days a year, and $0.7 \times ₹300 \div 0.6 \div 12 = ₹29$. When real data showed Ahmedabad costs only ₹27, we kept ₹29 as the floor so no city is priced below a sustainable minimum.

Replaying every year since 2000

An average price can hide disaster years. So we replayed every year from 2000 to 2024 as if we had sold each product that year at its cost price: claims = days paid × ₹300, premiums = price × 12. Above 100%, that year lost money.



Claims + premiums, year by year, for the three Basic products. Red bars are loss years. Solid line: 100%, break-even. Dashed line: the 60% target. Kavach Plus is not shown: it is flat at 60% every year, because every year hits the cap.

PRODUCT	COST PRICE	WORST YEAR	CLAIMS ÷ PREMIUMS	LOSS YEARS
Plus · Mumbai · cap 5	₹208	2010: 19 days, 5 paid	60%	0 of 25
Plus · Mumbai · cap 3	₹125	2010: 19 days, 3 paid	60%	0 of 25
Basic · Mumbai rain	₹45	2019: 4 days, 4 paid	222%	7 of 25
Basic · Nagpur heat	₹52	2013: 14 days, 5 paid	242%	5 of 25
Basic · Ahmedabad heat	₹27	2018: 7 days, 5 paid	469%	4 of 25

What it tells us

KAVACH PLUS NEVER LOST MONEY

Mumbai always has at least 5 heavy-rain days, so the payout is the same ₹1,500 every year. For a delivery app it is a **predictable rain allowance** that lands on the right days for the right riders.

KAVACH BASIC NEEDS REINSURANCE

Rare-event cover has bad years: 2 to 5 times the premiums. That is normal insurance. A reinsurer covers those years, and pooling cities helps because Mumbai's rain and Nagpur's heat peak in different months.

THE WORST SINGLE DAY

If a city-wide flood triggers every rider at once: 50,000 riders × ₹300 = **₹1.5 crore** in one day. The cap means it can happen at most 5 times a year per rider, so the yearly maximum is known: ₹7.5 crore for 50,000 riders.

Premiums, payouts and what Jio earns

The most important distinction: **premiums are not Jio's revenue**. A licensed insurer collects them and carries the risk. Jio earns a distribution fee (15–25% of premiums) and sells weather data.

Kavach Plus · one delivery app

Riders covered	50,000
Premiums: $50,000 \times ₹208 \times 12$	₹12.5 Cr
Back to riders (60%)	₹7.5 Cr
Jio's fees (15–25%)	₹1.9–3.1 Cr

Kavach Basic · at recharge

Jio users $\times 1\%$ who buy	50 crore $\times 1\% = 50$ lakh
Premiums: $50 \text{ lakh} \times ₹29 \times 12$	₹174 Cr
Back to people (60%)	₹104 Cr
Jio's fees (15–25%)	₹26–44 Cr

What moves the numbers

RIDERS	PREMIUMS ₹ CR	CLAIMS ₹ CR	JIO'S FEES ₹ CR
25,000	6.2	3.8	0.9–1.6
50,000	12.5	7.5	1.9–3.1
100,000	25.0	15.0	3.8–6.2
200,000	50.0	30.0	7.5–12.5

ADOPTION	POLICIES (LAKH)	PREMIUMS ₹ CR	JIO'S FEES ₹ CR
0.5%	25	87	13–22
1.0%	50	174	26–44
2.0%	100	348	52–87
5.0%	250	870	130–218

Basic uses the ₹29 floor price, the cautious case: city prices run up to ₹52. Adoption figures are illustrations of scale, not forecasts. 50 crore is an approximate count of Jio users.

A small, concrete example: Ahmedabad

1 lakh customers $\times ₹29 \times 12$	₹3.48 Cr premiums a year
Expected claims: $1 \text{ lakh} \times 0.64 \text{ days} \times ₹300$	₹1.92 Cr
Left for insurer, reinsurance, Jio's fee and costs	₹1.56 Cr

SAY IT LIKE THIS

"₹174 crore is the premium pool, not our revenue. Jio's share is 15 to 25 percent, the insurer carries the risk, and the data business comes on top."

Why we price the way we do

B2B

Riders first, sold to the app. A delivery app buys Kavach Plus for its riders. One contract covers tens of thousands of people, the app already knows where each rider works, and it has a clear reason to pay: storms cost it orders and riders. Individuals come later, at recharge.

₹7

Small per rider. ₹208 a month is about ₹7 a day, roughly 1% of a full-time rider's earnings. The 3-day version (₹125) gives the app a cheaper entry point.

City

Priced city by city, from data. A flat national price would overcharge low-risk cities and underprice high-risk ones. Each city's price comes from its own 25 years, with ₹29 as the floor.

Cap

A cap that makes the worst case known. At most 5 paid days a year: no one gets more than ₹1,500, and the insurer can reinsure a known maximum.

3 h

Daily history now, 3-hour triggers later. Indian records are daily, so prices come from daily data. The live product triggers on 3-hour rain. During the shadow pilot we record real 3-hour rain and the actuaries recalibrate the trigger to about 4–5 payout days a year before money moves.

Questions a judge may ask

Why 60%?

It leaves room for the insurer, reinsurance and our fee. Mass micro-insurance often runs lean on costs because distribution is digital, so a higher payout share is possible later.

Plus always pays its cap. Is that insurance?

For the app it works as an automated rain allowance: the money lands on the right days for the right riders, with no claims. Basic is the true insurance product.

What if a year is far worse than the last 25?

The cap bounds each person's payout, reinsurance covers the pool, and prices are reviewed every year as new data arrives.

Who sets the final price?

A licensed insurer's actuaries, under IRDAI rules. Ours is the evidence-based starting point.

What this does not prove, and what the pilot measures

Saying what we have *not* shown is what makes the rest believable. Each limit has a matching measurement in Phase 1.

LIMIT	WHY IT MATTERS	WHAT THE PILOT MEASURES
German links, not Jio's	Different equipment, frequencies and climate	The same test on Jio's Mumbai links against gauges and IMD radar
11 days in May	Not a monsoon, not a full season	A full 90-day monsoon, every storm
Radar is not perfect truth	Radar has its own errors	Comparison with Mumbai's automatic rain gauges as well
18% under-reading	Fixed triggers need correct amounts	Totals within $\pm 10\%$ of gauges after local calibration
68% of events caught	Missed payouts hurt trust	80%+ caught with fusion, under 15% false triggers
Midpoint map	Ignores that a link measures a line	Path-aware mapping; error per square by link length
Fibre replaces microwave in cities	Fewer links where we need density	Coverage audit: usable links per 1 km square
Daily data for 3-hour triggers	Prices may shift when triggers change	3-hour trigger counts; recalibration before money moves
Archive under-records extremes	Totals on the worst days are too low	Prices use counts of days above a threshold, which are robust
Adoption and app ROI unknown	The ₹174 Cr is scale, not a forecast	Orders lost and rider churn in the partner app's own data

The go / no-go scorecard for real payouts

$\geq 80\%$ heavy-rain events caught	$< 15\%$ false triggers	$\pm 10\%$ rain totals vs gauges	$< 6\text{ h}$ from storm to payout
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SAY IT LIKE THIS

"We proved the method on real network data and priced the product on 25 years of IMD data. What we haven't proved is Jio's own coverage in Mumbai, and that is exactly what our 90-day shadow pilot measures, before any money moves."

Three ways to re-check every number

The Test Lab

Open `JioMausam_Test_Lab_standalone.html`. It re-runs all 500 links from the raw signal in a few seconds and shows "✓ Matches our Python run". Move the sliders to test assumptions live.

The Excel workbook

`JioMausam_Numbers_Workbook.xlsx`: live formulas for K, the 0.95, the 0.88, the payout test, prices, the stress test and the business case. Change a yellow cell and watch it update.

The code

`rain_pipeline.py`, `india_layer.py`, `stress_test.py` in the repository's `prototype/` folder, plus the data pack's CSV files of every stage.

Glossary

Attenuation. Loss of signal strength, in dB. Rain attenuation is the part caused by rain in the path.

Baseline. The signal loss a link would have if it were dry right now.

Bias. Our total rain ÷ radar's total. 0.82 means we read 18% low.

Calibration (K). One multiplier that matches our totals to radar's on the tuning days.

Cap. The most days paid per person per year (5).

Correlation (r). How closely two series rise and fall together, from -1 to 1.

dB, dBm. Log scale for signal power; every 10 dB is ten times less. dBm is measured against 1 milliwatt.

Events caught. Share of radar's payout events our system also triggered (68%).

False trigger. We pay, radar would not have (18% of our triggers).

GSOD. NOAA's Global Summary of the Day: archived daily weather-station reports, including IMD's.

IDW. Inverse distance weighting: nearer links count more when filling a square.

ITU-R P.838. The international standard linking rain rate to microwave loss.

Loss ratio (payout share). Claims ÷ premiums. We price for 60%.

Microwave link (CML). A radio beam between two towers carrying network traffic.

Parametric insurance. Pays automatically when a measured value crosses a trigger; no claim forms.

pycomlink. KIT's open-source toolbox for rain from microwave links; source of our German data.

RADOLAN. The German Weather Service's radar rainfall product, our answer key.

Reinsurance. Insurance for the insurer, covering years when claims beat premiums.

Shadow mode. The system decides who it would pay, but no money moves.

Trigger. The measured level that causes a payout (10 mm in 3 hours; 64.5 mm a day; 47°C).

Tuning / test days. 10–12 May to fit K; 13–20 May, never seen in tuning, for every score.

Wet-antenna attenuation. Extra loss from water on the dishes, removed before computing rain.